



HCA 5110 Applied Healthcare Statistics

Final Project Report: Weekly Hospital Beds Occupancy Data

AARON WHETSTONE, NARJIS JEBALI, AND TIFFANY VOGLER

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I. Introduction

The objective of this report is to examine regional weekly hospital bed occupancy rates. We will report COVID-19, influenza, and Respiratory Syncytial Virus (RSV) trends along with overall bed occupancy rates across regions. Our dataset, "Weekly Hospital Beds Occupancy – Regional Level," contains detailed data on inpatient and ICU bed occupancy and hospitalization by disease and geography. These trends can be used to guide infrastructure planning, resource planning, and public health preparedness. Hospitals can make use of such information to make their staff better in different months, procure essential materials or medication ahead of time to adopt preventive measures. Governments at the national and local levels can also take informed decisions about the opening of new hospitals or upgrading existing hospitals to meet future needs. Hospitalization due to specific diseases can be used to inform public health interventions, for example, immunization campaigns and outbreak response policies.

II. Problem Description

This study aims to explore trends, relationships, and predictive patterns in hospital capacity and system stress across different regions and time periods. Through data visualization, hypothesis testing, and regression analysis, the study seeks to provide a deeper understanding of key factors affecting hospital occupancy and resource utilization.

The data visualization component examines the trends of COVID-19, influenza, and RSV across various regions, as well as the variation in inpatient bed occupancy over time, including a comparison of pre- and post-COVID periods. Hypothesis testing is conducted to determine whether hospitals with higher total new RSV admissions experience significantly higher inpatient bed occupancy rates and whether influenza admissions differ significantly between weeks in 2024. Finally, regression analysis is employed to assess whether inpatient bed occupancy can be accurately predicted based on total new admissions for COVID-19, flu, and RSV, along with region and time. Additionally, it investigates whether changes in RSV admissions among adult patients significantly impacts the number of occupied inpatient beds.

By addressing these questions, the study aims to offer valuable insights into hospital system dynamics, aiding in better resource planning and response strategies.

III. Data Analysis

Our original csv file contained 15,813 rows and 190 columns. The columns include week ending dates, geographic aggregation, and a variety of numerical values. The raw data included both state level data and regional data. For conciseness we decided to omit the state level data to focus on the region level values. There are 10 regions that represent the following geographic areas: (the map of the regions would be useful in the presentation)

- 1 – Connecticut, Maine, Massachusetts, New Hampshire, Rhode Island, and Vermont
- 2 – New Jersey, New York, Puerto Rico, and the Virgin Islands
- 3 – Delaware, DC, Maryland, Pennsylvania, Virginia, and West Virginia
- 4 – Alabama, Florida, Georgia, Kentucky, Mississippi, North Carolina, South Carolina, and Tennessee
- 5 – Illinois, Indiana, Michigan, Minnesota, Ohio, and Wisconsin
- 6 – Arkansas, Louisiana, New Mexico, Oklahoma, and Texas
- 7 – Iowa, Kansas, Missouri, and Nebraska
- 8 – Colorado, Montana, North Dakota, South Dakota, Utah, and Wyoming
- 9 – Arizona, California, Hawaii, Nevada, American Samoa, Commonwealth of the Northern Mariana Islands, Federated States of Micronesia, Guam, Marshall Islands, and Republic of Palau
- 10 – Alaska, Idaho, Oregon, and Washington

iv. Data Cleaning

Given the nature of the data set, we performed several cleaning procedures to focus on the most relevant information. First dropped columns that would not be used in our analysis. This reduced the dataset to 9 essential columns which helped focus our data set. Then we identified missing values particularly for RSV and influenza hospitalizations. There were 47 missing records for RSV and 1,719 for influenza. These missing values were noted during the analysis. We also identified and eliminated outliers like a particularly high inpatient bed occupancy (63,599,266), which was likely due to a data entry error. Cleaning our data helped our report and analysis become more reliable and accurate.

About data:

The dataset provides an overview of hospital capacity and patient hospitalizations, with a focus on COVID-19, influenza, and Respiratory Syncytial Virus (RSV). The time range of the data spans from August 8, 2020, to February 8, 2025, with the median and mean centered around November 8, 2022. This suggests that the dataset captures a critical period during and after the COVID-19 pandemic. It also offers insights into hospital resource utilization and patient hospitalization trends. The interquartile range for the week ending dates, from September 23, 2021, to December 24, 2023, highlights the dataset's focus on the pandemic and post-pandemic phases.

Hospital capacity metrics reveal significant variability in the size and resources of the facilities included in the dataset. The number of inpatient beds ranges from a minimum of 1,462 to a maximum of 224,956, with a median of 65,897 and a mean of 67,325. This indicates a relatively symmetric distribution. However, the number of ICU beds shows a right-skewed distribution, with a median of 9,344 and a mean of 10,123, suggesting that a small number of hospitals have disproportionately high ICU bed counts. Similarly, the occupancy rates for inpatient and ICU beds are right-skewed, with median occupied inpatient beds at 50,490 (mean: 76,917) and median occupied ICU beds at 6,078 (mean: 7,195). The maximum value for occupied inpatient beds (63,599,266) appears to be an outlier. This means we may need to investigate the accuracy of the data.

Hospitalization trends for COVID-19, influenza, and RSV reveal distinct patterns. COVID-19 hospitalizations exhibit significant variability, with a median of 1,483.5 and a mean of 3,258.7, indicating

periods of very high hospitalization rates that skew the distribution. Influenza hospitalizations are considerably lower, with a median of 66 and a mean of 265.2. There are quite a few missing values which means the data is not entirely complete. RSV hospitalizations are even lower, with a median of 4 and a mean of 116.4. There is an extreme number of missing values meaning there is not a lot of data available for RSV. These findings highlight the disproportionate impact of COVID-19 compared to influenza and RSV, as well as the need for more comprehensive data collection for the latter two respiratory illnesses.

In conclusion, the dataset provides valuable insights into hospital capacity and patient hospitalization trends during and after the COVID-19 pandemic. However, the presence of potential outliers, such as the extremely high value for occupied inpatient beds, and the significant number of missing values for influenza and RSV hospitalizations, downplays the importance of data quality checks and improved data collection practices.

Future analyses could focus on temporal trends, geographic disparities, and the impact of public health interventions on hospitalization rates for COVID-19, influenza, and RSV. Addressing these limitations would enhance the dataset's utility for informing healthcare policy and resource allocation decisions.

v. Descriptive Statistics

Total Patients Hospitalized with RSV

Statistic	Value
Min.	0.0
1st Qu.	0.0
Median	4.0
Mean	116.4
3rd Qu.	33.0
Max.	1895.0
NA's	1719

Total Patients Hospitalized with COVID-19

Statistic	Value
Min.	16.0
1st Qu.	662.8
Median	1483.5
Mean	3258.7
3rd Qu.	3608.2
Max.	43788.0

Total Patients Hospitalized with Influenza

Statistic	Value
Min.	0.0
1st Qu.	18.0
Median	66.0

Mean	265.2
3rd Qu.	199.0
Max.	7634.0
NA's	47

Number of Inpatient Beds Occupied

Statistic	Value
Min.	1043
1st Qu.	23826
Median	50490
Mean	76917
3rd Qu.	68174
Max.	63599266

Number of ICU Beds Occupied

Statistic	Value
Min.	59
1st Qu.	2860
Median	6078
Mean	7195
3rd Qu.	10268
Max.	26495

Number of Inpatient Beds

Statistic	Value
Min.	1462
1st Qu.	31452
Median	65897
Mean	67325
3rd Qu.	89245
Max.	224956

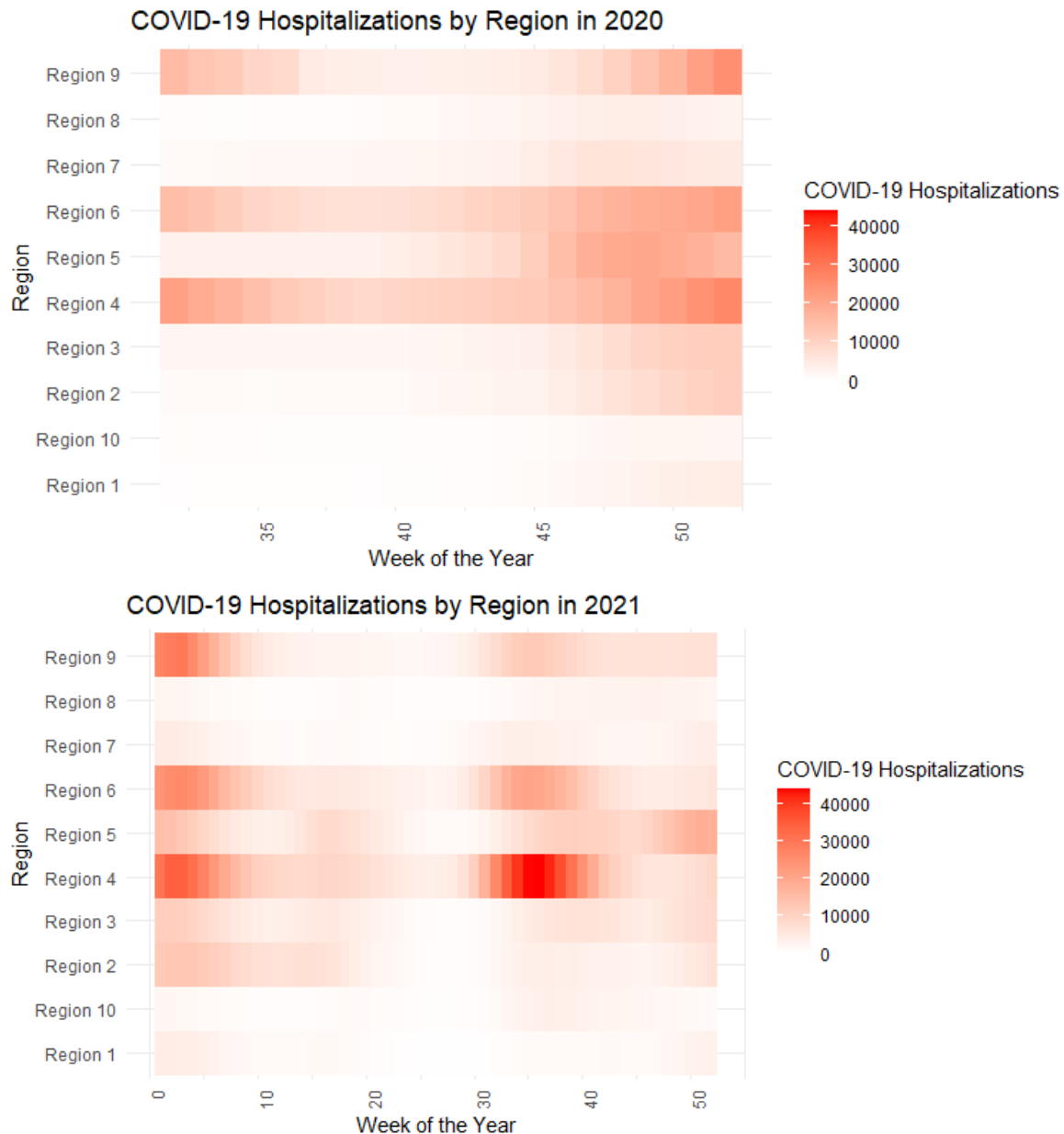
Number of ICU Beds

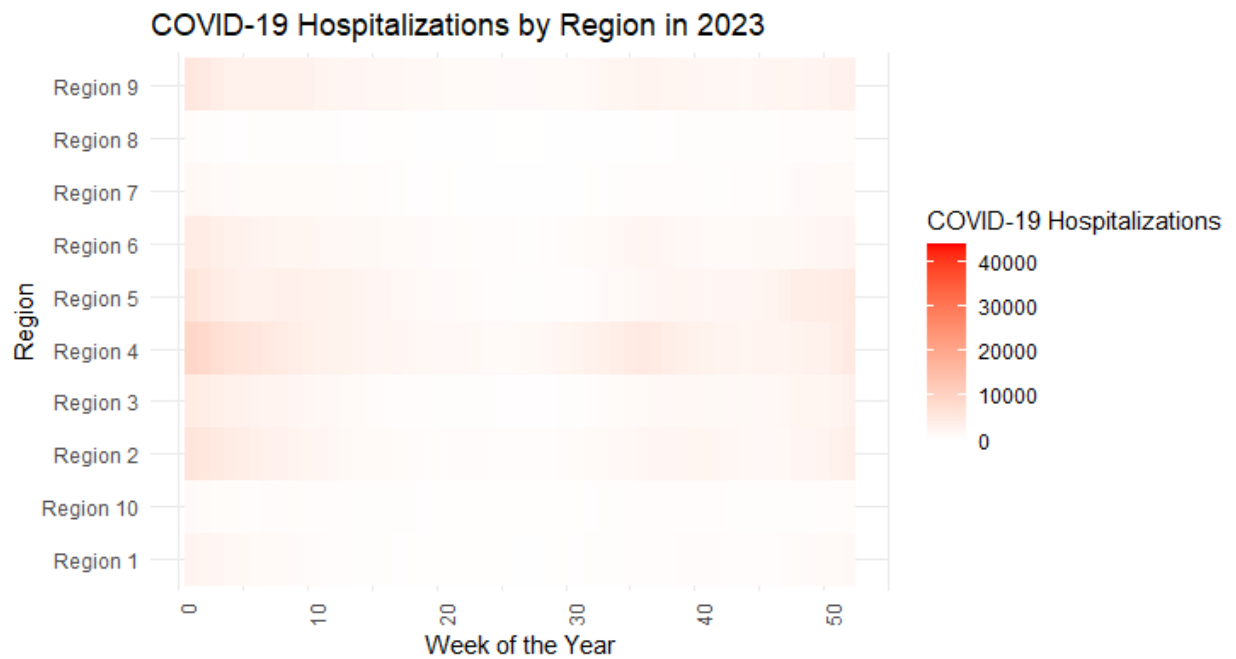
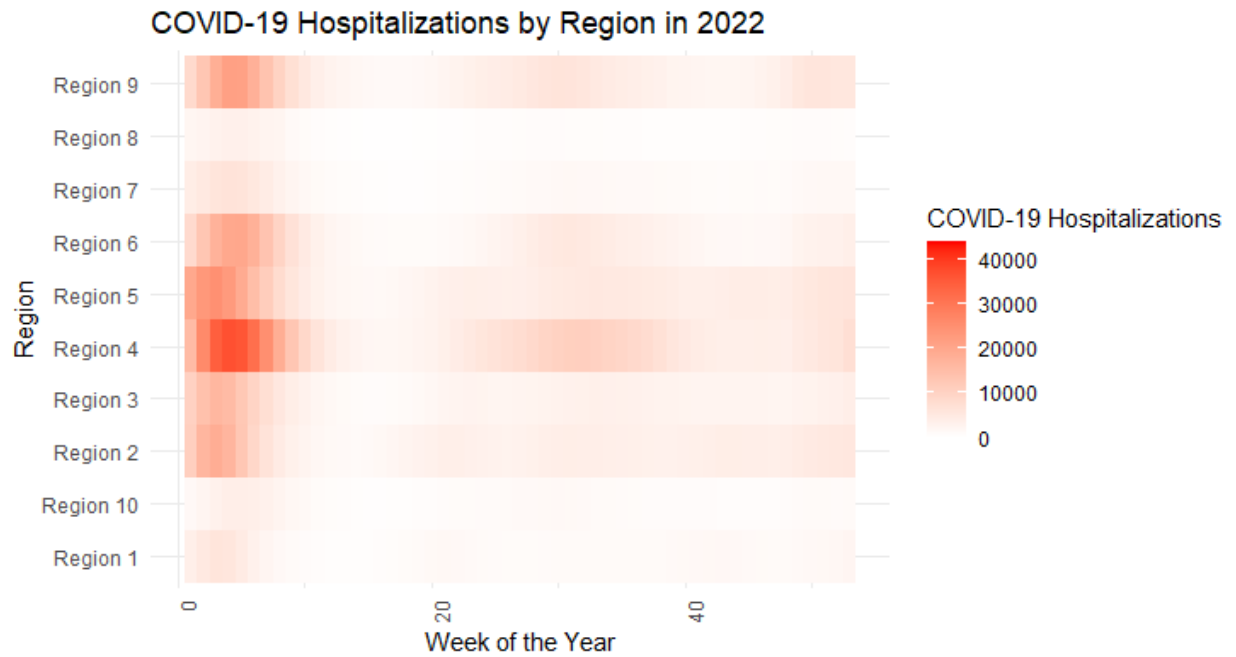
Statistic	Value
Min.	97
1st Qu.	4029
Median	9344
Mean	10123
3rd Qu.	14457
Max.	30765

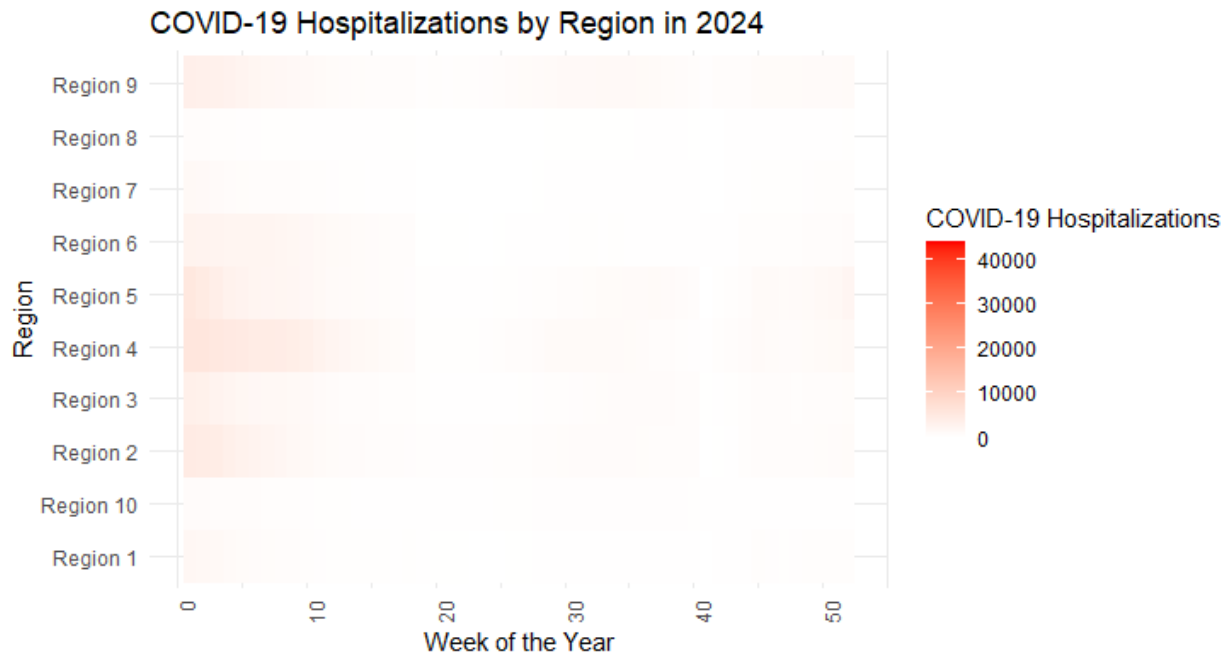
Both RSV and influenza hospitalizations have missing values with RSV having the most. We clean the data to remove those missing values prior to testing our hypotheses.

vi. Visualizations

Our data set offered the opportunity to come up with multiple visualizations that would help put the data into perspective. First, we created heat maps breaking down COVID-19 hospitalizations for each year broken down by region:

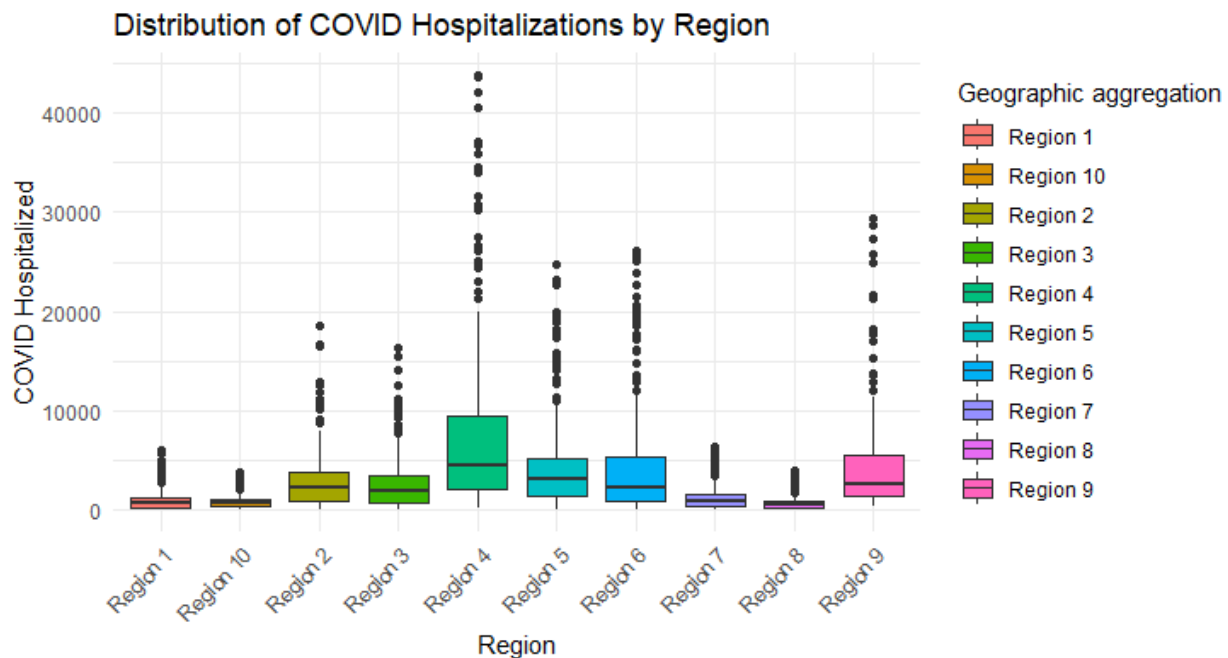






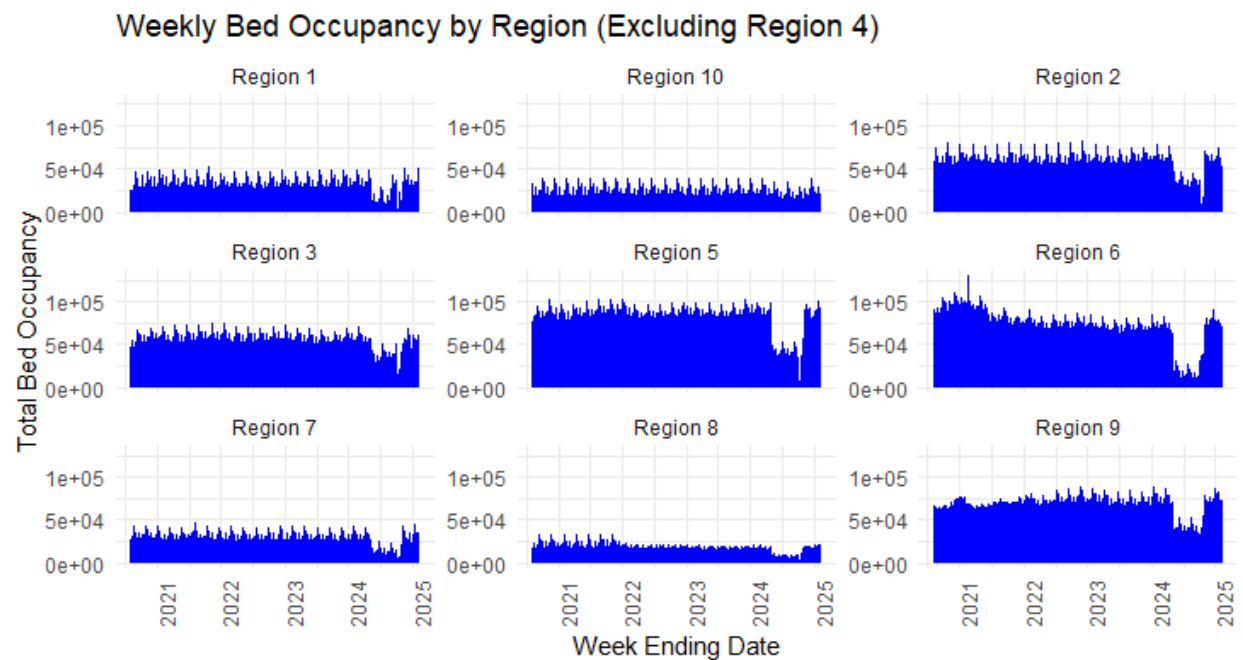
We made sure that the scale was the same each year to give the visualizations consistency. Region 4 had the largest spike but by the end of 2024 there were no regions with an abundance of COVID-19 hospitalizations.

The second visualization we created pertained to the overall distribution of COVID-19 hospitalizations broken down by region:



The box and whisker plot shows how region 4 has outliers that are skewing the data which can lead to problems as we analyze the data. For our final visualization we decided to remove region 4 since we

found that there is an outlier so egregious that it is probable that it was a keying error when entering the database.



Most regions in our data set experienced a severe dip in bed occupancy from the end of 2024 into 2025 outside of region 5 which stayed steady. As COVID-19 led to less hospitalizations, overall bed occupancy declined but there is still another spike in 2025. This report will analyze certain trends and information that may help answer how to fix this spike or future ones.

vii. Hypothesis Testing

Do hospitals that report higher total new RSV admissions also have a significantly higher inpatient bed occupancy rate?

To determine whether hospitals that report higher total new RSV admissions also have a significantly higher inpatient bed occupancy rate, an independent t-test was conducted. The analysis compared two distinct groups: hospitals with high RSV admissions and hospitals with low RSV admissions.

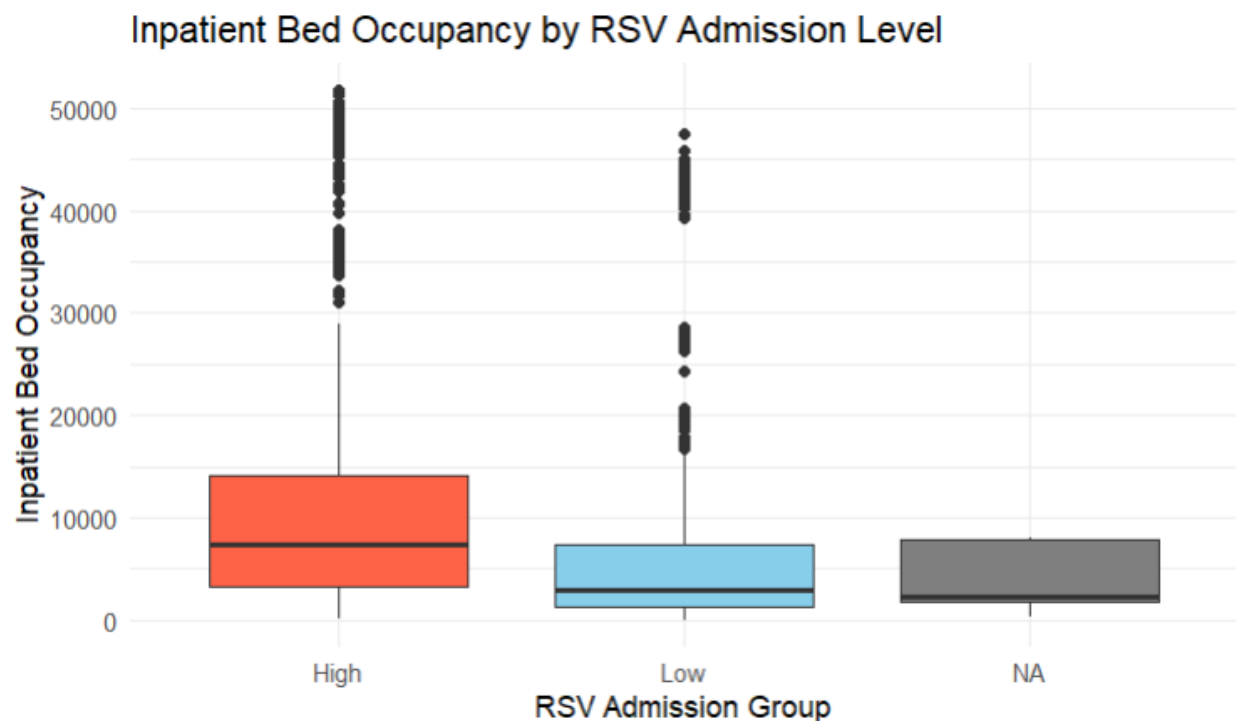
The null hypothesis (H_0) stated that there would be no difference in the mean inpatient bed occupancy rate between hospitals with high and low RSV admissions. Conversely, the alternative hypothesis (H_1) proposed that hospitals with higher RSV admissions would exhibit significantly different inpatient bed occupancy rates compared to those with lower admissions.

The results of the t-test are summarized in the table below:

Table 1: Independent t-test Results for Inpatient Bed Occupancy by RSV Admissions

Statistic	Value
t-value	13.08
Degrees of Freedom	1,949.4
p-value	< 2.2e-16
Mean (High RSV)	11,177.34
Mean (Low RSV)	5,874.82
95% Confidence Interval	(4,507.470, 6,097.577)

Given the exceptionally low p-value, the null hypothesis was rejected. These findings provide strong statistical evidence that hospitals with higher RSV admissions experience significantly higher inpatient bed occupancy rates compared to those with lower RSV admissions. The substantial difference in means underscores the considerable burden that RSV-related cases place on hospital resources, further emphasizing the need for capacity planning and effective management during peak RSV seasons.



Boxplot for RSV Admissions vs. Inpatient Bed Occupancy

The boxplot depicting inpatient bed occupancy rates by RSV admission levels provides a clear visualization of the differences identified in the t-test. Hospitals with high RSV admissions exhibited a notably higher median inpatient bed occupancy compared to those with low RSV admissions. Additionally, the high RSV admissions group displayed a greater spread of data, with several extreme

outliers, indicating that some hospitals experience exceedingly high inpatient occupancy due to RSV-related cases.

In contrast, hospitals in the low RSV admissions group had a significantly lower median occupancy, though a few outliers suggested occasional spikes in occupancy levels. If the "NA" category refers to hospitals that did not report RSV admissions, these facilities tended to have the lowest inpatient bed occupancy.

This visualization supports the statistical findings, reinforcing the conclusion that hospitals experiencing high RSV admissions tend to face significantly greater occupancy challenges. The presence of outliers also suggests that certain hospitals bear a disproportionate burden, potentially due to localized outbreaks or variations in healthcare capacity.

Is there a significant difference in the number of influenza admissions between different weeks of the year 2024?

To assess whether the number of influenza admissions varied significantly across different weeks in 2024, an Analysis of Variance (ANOVA) test was performed. Given that the study involved multiple weeks, ANOVA was the most appropriate statistical approach for evaluating potential differences in means over time.

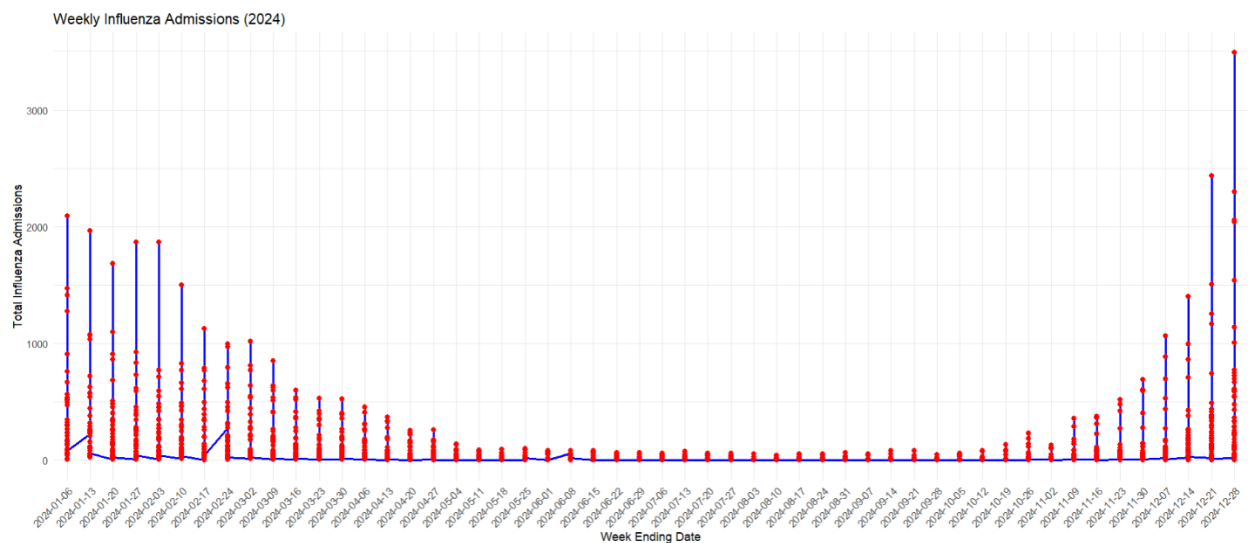
The null hypothesis (H_0) posited that there would be no significant difference in the number of influenza admissions across the various weeks, indicating a consistent trend throughout the year. The alternative hypothesis (H_1), on the other hand, suggested that influenza admissions fluctuated significantly across different weeks.

The ANOVA test results are presented in the table below:

Table 2: ANOVA Results for Weekly Influenza Admissions

Source of Variation	Degrees of Freedom (df)	F-value	p-value
Between Weeks	51	14.66	< 2e-16
Within Groups	2,121	-	-

As a result, the null hypothesis was rejected, confirming that influenza admissions varied significantly across different weeks in 2024. These results suggest the presence of seasonal trends, outbreaks, or other external factors influencing influenza hospitalizations throughout the year. The significant fluctuations further highlight the need for hospitals to anticipate periods of increased patient influx and allocate resources accordingly.



Line Plot for Weekly Influenza Admissions

The line and scatter plot illustrating weekly influenza admissions in 2024 further substantiates the results of the ANOVA test. The visualization reveals distinct seasonal trends in influenza cases, with notable peaks occurring in January and December. The highest recorded admissions were observed in early January and late December, aligning with well-documented influenza seasonality patterns.

Conversely, influenza admissions experienced a significant decline during the mid-year months, particularly in summer, before gradually rising again in the latter part of the year. The red data points represent individual hospital admissions, while the blue line connects weekly averages, providing a smoothed representation of the overall trend.

These observations align with the statistical findings, confirming that influenza admissions vary considerably across different weeks. The marked fluctuations suggest that hospitals must prepare for sharp increases in patient volume during peak influenza seasons, ensuring adequate staffing, bed availability, and medical supplies.

To conclude this part, the hypothesis testing confirms two key findings. First, hospitals experiencing higher RSV admissions also face significantly higher inpatient bed occupancy rates, as demonstrated through both statistical testing and visual representation. The substantial difference in mean occupancy rates underscores the strain placed on hospital capacity by RSV-related cases. Second, influenza admissions exhibit significant weekly variations throughout 2024, with pronounced seasonal trends characterized by winter peaks and summer declines. The ANOVA test and corresponding visualizations strongly support this pattern, highlighting the importance of resource allocation and planning for influenza surges.

Together, these findings emphasize the need for data-driven hospital management strategies to ensure that facilities are adequately prepared for RSV and influenza-related fluctuations. Understanding these trends can aid in optimizing patient care, mitigating resource shortages, and improving overall healthcare system efficiency.

VIII. Regression Analysis

We came up with two questions to explore during our regression analysis:

Can the number of occupied inpatient beds be accurately predicted using the total reported new admissions for COVID-19, flu, and RSV, along with region and time (week)?

Is the number of occupied inpatient beds significantly affected by changes in RSV admissions among adult patients?

We constructed a multiple regression analysis model to examine the relationship between bed occupancy and factors including COVID-19, flu, and RSV hospitalizations including any outliers. Our results show that only COVID-19 hospitalizations had a p-value < 0.001 meaning it was statistically significant. Flu and RSV hospitalizations are not statistically significant with bed occupancy levels. Overall, our model has an R-square value of 0.01993 meaning it predicts less than 2% of the variance in inpatient bed occupancy. That is a very poor fit, meaning that in its current form our model is not very useful.

We then identified and removed outliers in region 4 that skewed the results. Our new model was significantly improved. Here are the results:

<u>Variable</u>	<u>Estimate</u>	<u>Std. Error</u>	<u>t-value</u>	<u>p-value</u>
Intercept	2.409e+05	1.168e+04	20.624	(< 2e-16)
COVID_Hospitalized	9.611e-01	6.128e-02	15.683	(< 2e-16)
Flu_Hospitalized	6.344e+00	5.394e-01	11.761	(< 2e-16)
RSV_Hospitalized	4.126e+00	2.002e+00	2.061	(0.0394)
Region 10	-7.624e+03	1.068e+03	-7.142	(1.22e-12)
Region 2	2.318e+04	1.077e+03	21.513	(< 2e-16)
Region 3	2.105e+04	1.074e+03	19.593	(< 2e-16)
Region 4	8.232e+04	1.169e+03	70.445	(< 2e-16)
Region 5	4.529e+04	1.099e+03	41.206	(< 2e-16)
Region 6	3.611e+04	1.096e+03	32.955	(< 2e-16)
Region 7	-2.028e+03	1.068e+03	-1.899	(0.0576)
Region 8	-9.317e+03	1.068e+03	-8.726	(< 2e-16)

Region 9	3.370e+04	1.093e+03	30.833	(< 2e-16)
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Region 7 is the only variable that has a p-value > 0.05. We also ran the model fit statistics and here are the results:

Metric	Value
Residual Standard Error	11,590
Multiple R-squared	0.8815
Adjusted R-squared	0.8808
p-value	< 2.2e-16

We completed a multicollinearity check, and our results indicated no significant multicollinearity. Removing the outliers in region 4 greatly improved our model.

ix. Key Findings

We identified three findings from our analysis:

1. COVID-19, Influenza, and RSV hospitalizations significantly impact inpatient bed occupancy.
 - a. Hospitals are more burdened by flu and RSV patients than by COVID patients.
2. Significant Regional Differences in Bed Occupancy
 - a. Resource allocation should be region-specific, focusing on areas with the highest burden.
3. Seasonal Trend – Occupancy is Declining Over Time
 - a. The model is statistically robust, meaning these factors (hospitalizations, region, time) accurately predict occupancy.

x. Future Work

Future work on this subject can include exploring some of the key limitations and other areas of hospital resource utilization. Researchers could examine how other respiratory illnesses impact bed use rates to better understand seasonal healthcare demand. Also adding information on the level of staffing in hospitals, supply chain logistics, and patient outcomes may provide further insights into operational hurdles that hospitals experience. Predictive modeling using historical data to predict future trends in hospitalizations can help hospitals be more proactive instead of reactive in how they handle many aspects of their operation. Comparative studies between countries or healthcare systems could determine best practices for managing bed occupancy and improved patient care during public health emergencies. By bridging such gaps, follow-up research would be able to offer insights into more resilient and more effective healthcare systems that can better cope with both routine and exceptional demands.

Further analysis of this data should include percentage-based capacity versus raw bed numbers. This will more accurately reflect overall capacity availability and constraints. Resources can then be allocated appropriately. For example, staffing resources can be sent to the facilities with a greater number of

open beds to maximize capacity. Alternatively, patients could be transported from overburdened facilities to those that have a greater percentage of open beds.

xi. Conclusion

Our report analyzed weekly hospital bed utilization rates and disease-specific hospitalization trends in 10 geographic areas of the United States for COVID-19, influenza, and RSV. The data spans from August 2020 to February 2025 and provided insights into hospital resource utilization during and after the COVID-19 pandemic. Key findings revealed hospitalizations for COVID-19 were most impactful on bed occupancy with a median of 1,483.5 and a mean of 3,258.7. Hospitalizations for influenza and RSV, though less significant, also contributed to occupancy. Influenza showed seasonality while RSV data was limited by a high proportion of missing values. Regional variation in occupancy was apparent, with heavily populated regions such as Region 2 and Region 9 with higher occupancy rates than sparsely populated regions such as Region 8.

There were limitations and issues with our data set that we identified. Outliers like a high value for inpatient occupied beds 63,599,266 and missing values for influenza and RSV hospitalizations show how improved data collection is necessary. Fortunately, removing these outliers to complete data analysis leads to an impressive regression model that can predict almost 90% of the variability in hospital bed occupancy. Despite these limitations, the findings are informative for healthcare systems and policy makers. Hospitals can use these trends to optimize staffing, budget resources, and prepare for respiratory disease seasonals. The information we found can help healthcare officials make informed, evidence-based decisions about investing in infrastructure. This includes expanding hospital capacity or building new facilities where demand is highest based on trends.

Future research can help improve this study by incorporating other variables, such as staffing, supply chain operations, and patient outcomes. This will increase information to understand hospital resource use. Predictive modeling can be employed to forecast future hospitalization trends and anticipated resource planning. Once more knowledge gaps are closed, subsequent research can assist in planning more resilient and efficient healthcare systems.

In conclusion, this report illustrates trends in hospital bed occupancy for a variety of diseases including COVID-19. By looking at the trends we identified, stakeholders will be better poised to address challenges that the healthcare system faces.

xii. Reference

HealthData.gov. (n.d.). *Weekly hospital respiratory data (HRD) metrics by jurisdiction*. U.S. Department of Health and Human Services. Retrieved March 10, 2025, from https://healthdata.gov/dataset/Weekly-Hospital-Respiratory-Data-HRD-Metrics-by-Ju/n3kj-exp9/about_data

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